

VISTA Seminar

Seminar 114

October 14, 2026

10:00 am – 11:30 am EDT Buffalo / 3:00 – 4:30 pm BST London / 4:00 pm – 5:30 pm CEST Paris / 10 pm – 11:30 pm CST Beijing

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Electrophore model for dissociation of molecules after electron impact in plasma

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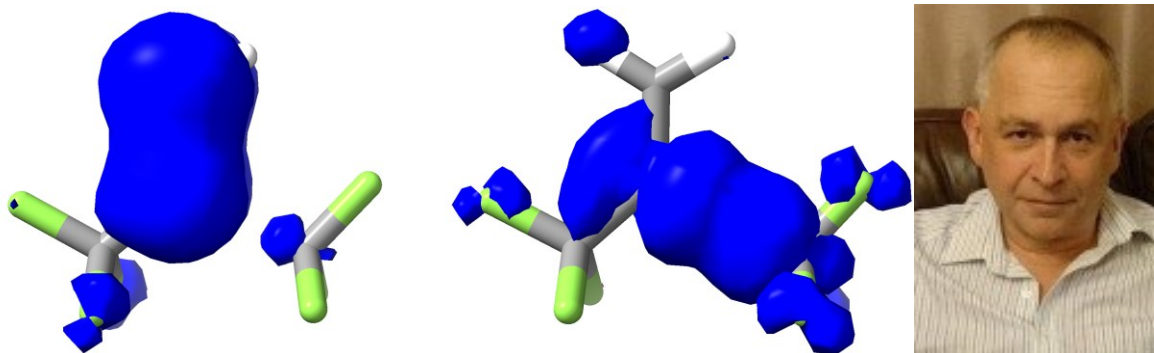
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Chemical processes in low-temperature plasmas typically involve molecules excited by electron impact. Unlike photoexcitation, electron impact on closed shell species primarily produces molecules in triplet states, as triplet states are lower in energy. However, similar to excitation of singlet states by a photon, triplet excitation is localized in a small part of the molecule, which we called electrophore, by analogy with chromophore in photochemistry.

The electrophore model suggested by us [1,2] describes well the dynamics of dissociation in plasma. We have shown that nonadiabatic dynamics of energy transfer between different triplet states can be modeled using theories and codes developed previously for the dynamics of energy transfer between chromophores in photochemistry. We find that molecules in triplet states created by electron impact relax very quickly to the lowest triplet state and the dynamics of dissociation simplifies. Therefore, *Ab initio* molecular dynamics on the lowest triplet Potential Energy Surface is often sufficient for understanding the reaction. Plasma chemistry is crucial for plasma technologies in microelectronics manufacturing, and it has been a subject of many experimental studies. In particular, industry is looking for new environmentally friendly molecules to replace fluorocarbons currently used in plasma etching and plasma deposition. See for example [3,4]. Almost no theory of dissociation after electron impact has been reported previously. We hope that this work initiates a new direction of research for the nonadiabatic dynamics community and will help to find new molecules for the use in industry.

References

- [1] Makhov, D.V., Armstrong, G., Chuang, H.H., Ambalampitiya, H., Lemishko, K., Mohr, S., Nelson, A., Tennyson, J., and Shalashilin, D.V. *Journal of Physical Chemistry Letters*, **2024** **15**, 3404–3411.
- [2] Brook, R., Bramley, O., Makhov, D.V., Nelson, A., Armstrong, G., Yong, J., Saunders, E., de Viggiani, J., Tennyson, J., Shalashilin, D.V. *Journal of Vacuum Science and Technology* **2025** **A43**, 043003.
- [3] Lee, H.J., Tak, H.W., Kim, S.B., Kim, S.K., Park, T.H., Kim, J.Y., Sung, D., Lee, W., Lee, S.B., Kim, K., Cho, B.O., Kim, Y.L., Lee, K.C., Kim, D.W., and Yeom, G.Y. *Applied surface science*. **2023** **639**, 158190.
- [4] Kim, J., Kang, H., Kim, Y., Jeon, M., and Chae, H. *Plasma processes and polymers*. **2024** **21**, 5.

ON THE INTERPRETATION OF MOLECULAR PHOTOEXCITATION WITH LONG AND ULTRASHORT LASER PULSES

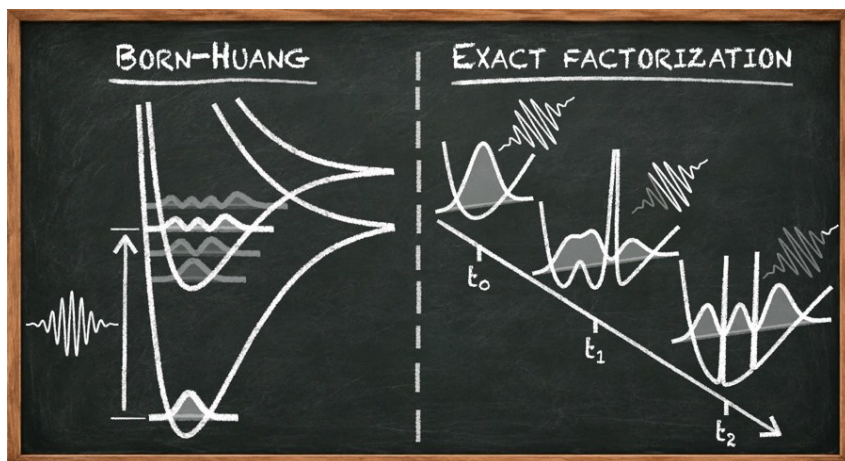
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Photoexcitation is an inherent part of any photochemical or spectroscopic experiment, yet its role in excited-state dynamics is often overlooked. In practice, however, it is the photoexcited molecular state, built upon photoexcitation and shaped by the characteristics of the light source, that ultimately determines the fate of the excited molecule and its subsequent photochemical behavior. In this talk, I will discuss how different laser pulses build excited molecular states, using two complementary representations of the molecular wave function: the Born–Huang expansion and exact factorization. I will consider two limiting cases: the preparation of a stationary molecular state by a long 100-fs laser pulse and the generation of an electronic wave packet by an ultrashort (attosecond) pulse. I will further show that standard concepts such as population transfer between electronic states, resonance conditions, and sudden vertical excitation—concepts naturally associated with the Born–Huang representation and widely used in chemical thinking about photoexcitation—are fundamentally challenged when viewed through the framework of exact factorization. Finally, I will comment on suitable strategies for modelling photoexcitation in both femto- and attochemistry based on the presented results.

How to connect

Alexey Akimov is inviting you to a scheduled Zoom meeting.

Topic: Seminar #114: Dmitry Shalashilin and Jiri Janos

Time: Oct 14, 2026 10:00 AM Eastern Time (US and Canada)

Join Zoom Meeting

<https://buffalo.zoom.us/j/94252964712?pwd=YlJ0rkx8naBvge59N9aTWZlu5w7MMJ.1>

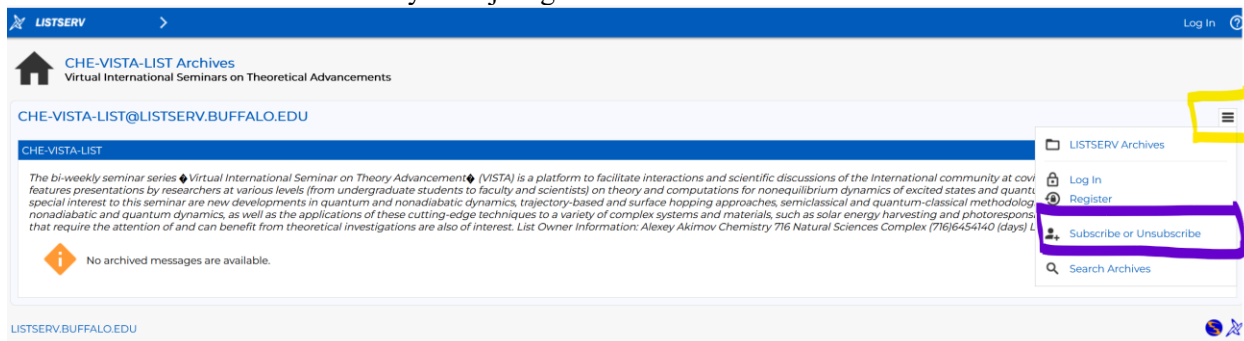
Meeting ID: 942 5296 4712

Passcode: 004303

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A. VISTA Mailing list:

1. Follow the link: <https://listserv.buffalo.edu/scripts/wa.exe?A0=CHE-VISTA-LIST&X=OA41BBB2DC6071987DF&Y=alexeyak%40buffalo.edu>
2. Click the menu icon in the upper right part of the list (yellow highlight in the picture below)
3. Click the “Subscribe or Unsubscribe” option (purple highlight below) – it will bring you to the next window where you’ll be asked for your email/name (I think it the name is optional to provide). This way, you can subscribe to the mailing list to stay tuned or unsubscribe if you find the seminars irrelevant to you or just get too much emails to deal with.



B. Slack Workspaces:

1. VISTA workspace: https://join.slack.com/t/vista-atk8254/shared_invite/zt-mdlteo5v-P1Hc7XVupkwMbnGhNG4KIw
2. Quantum Dynamics Hub workspace: https://join.slack.com/t/quantumdynamicshub/shared_invite/zt-mjbhjsxx-GGhsbYHxeBMvhmumK_j7LA

C. Gmail calendar:

<https://calendar.google.com/calendar/u/4?cid=Y19jMjhjZjc3YmQxZWY1MWZkMzAwNjQ2MDVkcZmQ1YjY3OGMyZmMxMGJmYjZhZmUyOGViZjg0MzA0NzVhMmY5NDAYQGdyb3VwLmNhbGVuZGFyLmdvb2dsZS5jb20>